الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

UNIT 2 · LESSONS 2-1 · 2-2 · 2-3

Preparation Booklet

Formative Assessment 1 — Retake
Grade 12 Advanced · Mathematics

كتيّب التجهّز للاختبار التكويني الأول (إعادة)

1 2 3 4 5 6 7

7	20	40	21
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Thursday 8 October 2026**

One page per objective · every solution written out step by step

Test map

7 questions = 7 recipes · 7 أسئلة = 7 وصفات

1 **Slope at a point** ميل المنحنى
Secant lines on both sides: $m = (y_2 - y_1) / (x_2 - x_1)$

2 **Arc length (n = 2)** طول المنحنى
3 points → 2 distances → add

3 **Estimate with a table** جدول القيم
x on both sides → f(x) → what does it approach?

4 **Graph or |...|** من جهة واحدة
Left, right, compare: equal? otherwise “does not exist”

5 **Algebraic limit (0/0)** جبريًا
Factor → cancel → substitute

6 **Root in the fraction · sin x / x** sin x/x · المرافق
Conjugate → cancel · or make the inside of sin = the denominator

7 **An inequality traps the function** نظرية الشطيرة
Bound → multiply → outer limits → “by the Squeeze Theorem”

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 The slope of a curve at a point

ميل المنحنى عند نقطة

The book's own exercises 1–6 are the model. The Ministry tests this idea through arc length.

The slope at a point is the number the secant slopes get closer and closer to.

ميل المنحنى عند نقطة = العدد الذي تقترب منه ميول القواطع.

Recipe الوصفة

1. **Find the point (a, f(a))** — أوجد النقطة
2. **Pick second points on both sides of a** — نقاط ثانية من الجهتين
3. **Compute m_sec = $(y_2 - y_1) / (x_2 - x_1)$** — ميل كل قاطع
4. **See what the slopes approach** — إلى ماذا تقترب؟

Watch out

- Use both sides of a.
- Subtract in the same order, top and bottom.

Worked example Book exercise 2(a)

Estimate the slope of $y = x^3 + 2$ at $x = 1$, using the second points $x = 1.1,\ 1.01,\ 0.9,\ 0.99$.

1. substitute x = 1 → the point

$$f(1) = 1^3 + 2 = 3 \Rightarrow (1,\ 3)$$

2. second points: put each x into the function

$$(1.1,\ 3.331),\ (1.01,\ 3.030301)$$

$$(0.9,\ 2.729),\ (0.99,\ 2.970299)$$

3. m_sec for each second point

$$\frac{0.331}{0.1} = 3.31, \quad \frac{0.030301}{0.01} = 3.0301$$

$$\frac{-0.271}{-0.1} = 2.71, \quad \frac{-0.029701}{-0.01} = 2.9701$$

4. slopes → 3 from both sides

$$\text{slope} \approx \boxed{3}$$

Your turn · solve in your notebook · solutions at the back

1. **Method of example 1.1** Estimate the slope of $y = x^2 - 3x$ at $x = 1$, using the second points $x = 1.1,\ 1.01,\ 0.9,\ 0.99$.
2. **Book exercise 2(b)** Estimate the slope of $y = x^3 + 2$ at $x = 2$, using the second points $x = 2.1,\ 2.01,\ 1.9,\ 1.99$.

2 The length of a curve

طول المنحنى

Ministry: arc length comes back every year since 2022/2023, always with n = 2 line segments.

The length of a curve ≈ the sum of the lengths of line segments joining points on it.

طول المنحنى ≈ مجموع أطوال القطع المستقيمة بين نقاط عليه.

Recipe الوصفة

1. **Split the interval into 2 equal pieces** — قسّم الفترة جزأين
2. **Find the 3 points on the curve** — ثلاث نقاط
3. **Distance formula for each segment** — المسافة لكل قطعة
4. **Add the two lengths** — اجمع

Watch out

- n = 2 means 2 segments: 3 points, 2 distances.
- Exact middle: for [0, 3] it is 1.5.

Worked example Ministry 2024/2025

Estimate the arc length of the curve $y = x^2 - 4$ on the interval $0 \le x \le 2$ using $n = 2$ line segments.

1. 3 x-values: start, middle, end

   $x = 0,\ 1,\ 2$

2. heights from $y = x^2 - 4$

   $(0, -4),\ (1, -3),\ (2, 0)$

3. length of segment 1

   $d_1 = \sqrt{(1-0)^2 + (-3+4)^2} = \sqrt{2}$

4. length of segment 2

   $d_2 = \sqrt{(2-1)^2 + (0+3)^2} = \sqrt{10}$

5. add the two lengths

   $L \approx \sqrt{2} + \sqrt{10} \approx \boxed{4.5765}$

Your turn · solve in your notebook · solutions at the back

1. Ministry 2023/2024 Estimate the arc length of the curve $y = (x-1)^2$ for $0 \le x \le 3$ using $n = 2$ line segments.
2. Ministry 2025/2026 Estimate the arc length of the curve $y = \sin x$ on the interval $0 \le x \le \pi$ using $n = 2$ line segments.
3. Formative Assessment 1 Estimate the length of the curve $y = x^2 + 1$ for $-2 \le x \le 2$ using $n = 2$ line segments.

3 The concept of a limit

مفهوم النهاية

Ministry: estimate a limit from a table of values (2024/2025 and 2023/2024).

The limit tells what f(x) does near a, not at a. f(a) may not even exist.

النهاية تصف سلوك f(x) قرب a، لا عند a.

Recipe الوصفة

1. **Choose x close to a, on both sides**
   x قريبة من الجهتين
2. **Build a table: x and f(x)**
   ابنِ جدولًا
3. **What does f(x) approach?**
   إلى ماذا تقترب f(x)؟
4. **Write lim ≈ ...**
   اكتب التقدير

Watch out

- One side is not enough: use both.
- Do not substitute x = a into 0/0.

Worked example Ministry 2024/2025

Estimate $\lim_{x\to 0}\dfrac{\sin x - x}{x^3}$ numerically.

1. $x \to 0^-$ (left side)

   $f(-0.1) \approx -0.166583, \quad f(-0.01) \approx -0.166666$

2. $x \to 0^+$ (right side)

   $f(0.01) \approx -0.166666, \quad f(0.1) \approx -0.166583$

3. the values approach -0.1667

   $$-0.1667 \approx -\frac{1}{6}$$

4. estimate (calculator in radians)

   $$\lim_{x\to 0}\frac{\sin x - x}{x^3} \approx \boxed{-\frac{1}{6}}$$

Your turn · solve in your notebook · solutions at the back

1. **Book exercise 3** Estimate $\lim_{x\to 2}\dfrac{x-2}{x^2-4}$ with a table: $x = 1.9,\ 1.99,\ 2.01,\ 2.1$.
2. **Method of example 2.1** Estimate $\lim_{x\to 1}\dfrac{x^3-1}{x-1}$ with a table: $x = 0.9,\ 0.99,\ 1.01,\ 1.1$.

4 One-sided limits and existence

النهايات من جهة واحدة ووجود النهاية

Lesson 2-2 · Part 2 · pp. 70–73
MAT.6.01.07.002

Ministry: reading a limit from a graph appeared in 6 different years, the most common limit item.

The limit exists only when both one-sided limits exist and are equal.

النهاية موجودة فقط إذا تساوت النهايتان من اليمين واليسار.

Recipe الوصفة

1. **Left-hand limit: $x \to a^-$** — من اليسار
2. **Right-hand limit: $x \to a^+$** — من اليمين
3. **Equal → exists. Different → does not exist (give both values)** — متساويتان؟ وإلا غير موجودة

Watch out

- "Does not exist" alone loses marks: give both values.
- Closed circle = value. Open circle = approached only.

Tool. $|u| = u$ if $u > 0$ · $|u| = -u$ if $u < 0$

Worked example Ministry 2018/2019

Evaluate $\lim_{x\to 2} \dfrac{x-2}{|x-2|}$, if it exists.

1. right side: $x > 2$, so $|x-2| = x-2$

$$\lim_{x\to 2^+} \frac{x-2}{x-2} = \boxed{1}$$

2. left side: $x < 2$, so $|x-2| = -(x-2)$

$$\lim_{x\to 2^-} \frac{x-2}{-(x-2)} = \boxed{-1}$$

3. compare the one-sided limits

$$1 \neq -1$$

4. different → does not exist

$$\lim_{x\to 2} \frac{x-2}{|x-2|} \text{ does not exist}$$

Your turn · solve in your notebook · solutions at the back

1. Ministry 2021/2022 Evaluate $\lim_{x\to -1^+} \dfrac{|x+1|}{x^2-1}$.
2. Ministry graph item Use the graph to find: (a) $\lim_{x\to -2^-} f(x)$, (b) $\lim_{x\to -2^+} f(x)$, (c) $\lim_{x\to -2} f(x)$, (d) $\lim_{x\to 1} f(x)$, (e) $f(1)$.
3. Ministry 2019/2020 (paper part) Find $\lim_{x\to 0} \dfrac{2x-|x|}{|3x|-2x}$, if it exists.

5 Find a limit algebraically

إيجاد النهاية جبريًا

Ministry: factoring limits (difference of cubes, quadratics) are in almost every paper.

If substitution gives 0/0, factor, cancel the common factor, then substitute.

إذا أعطى التعويض 0/0: حلّل، اختصر، ثم عوّض.

Recipe الوصفة

1. **Substitute first** — عوّض أولًا
2. **If 0/0: factor top and bottom** — حلّل البسط والمقام
3. **Cancel the common factor** — اختصر العامل المشترك
4. **Substitute again** — عوّض من جديد

Watch out

- 0/0 does not mean "does not exist".
- Cancel factors, never terms.

Tool. $a^2 - b^2 = (a - b)(a + b)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Worked example Ministry 2022/2023

Evaluate $\lim_{x \to 1} \dfrac{x^3 - 1}{x^2 + 2x - 3}$.

1. substitute: 0/0 → simplify

$$\frac{1 - 1}{1 + 2 - 3} = \frac{0}{0} \Rightarrow \text{simplify}$$

2. factor top and bottom

$$\lim_{x \to 1} \frac{(x - 1)(x^2 + x + 1)}{(x - 1)(x + 3)}$$

3. cancel (x − 1)

$$\lim_{x \to 1} \frac{x^2 + x + 1}{x + 3}$$

4. substitute x = 1

$$\frac{1 + 1 + 1}{1 + 3} = \boxed{\frac{3}{4}}$$

Your turn · solve in your notebook · solutions at the back

1. Ministry 2018/2019 (paper part) Find $\lim_{x \to 2} \dfrac{x^3 - 8}{2x^2 - 3x - 2}$, if it exists.
2. Ministry 2020/2021 Evaluate $\lim_{x \to 0} \dfrac{\frac{1}{x+2} - \frac{1}{2}}{x}$, if it exists.
3. Ministry 2019/2020 Find $\lim_{x \to 2} \left(\dfrac{1}{x - 2} - \dfrac{4}{x^2 - 4} \right)$, if it exists.

6 Theorems: rationalizing

النهايات بالنظريات: الضرب في المرافق

Ministry: the most frequent limit item, 13 times in 9 years, mostly with a square root.

0/0 with a root in the fraction: rationalize, then cancel.

0/0 وفي الكسر جذر: اضرب في المرافق ثم اختصر.

Recipe الوصفة

1. **Substitute: 0/0 with a root** — 0/0 وفيه جذر
2. **Rationalize: multiply top and bottom by the conjugate** — اضرب في المرافق
3. **Simplify: (a − b)(a + b) = a² − b²** — بسّط المقام
4. **Cancel and substitute** — اختصر وعوّض

Watch out

- Multiply top AND bottom by the same expression.
- Quotient law needs the denominator's limit ≠ 0.

Tool. Theorem 3.1: sum · difference · constant multiple · product · power · root · quotient

Worked example Ministry 2021/2022 · book exercise 14

Evaluate $\lim_{x\to 0} \dfrac{2x}{3-\sqrt{x+9}}$.

1. substitute: 0/0 with a root

$$\frac{0}{3-3} = \frac{0}{0} \Rightarrow \text{rationalize}$$

2. rationalize the denominator

$$\lim_{x\to 0} \frac{2x\,(3+\sqrt{x+9})}{(3-\sqrt{x+9})(3+\sqrt{x+9})}$$

3. $(a-b)(a+b) = a^2 - b^2$

$$\lim_{x\to 0} \frac{2x\,(3+\sqrt{x+9})}{-x}$$

4. cancel x

$$\lim_{x\to 0} \left[-2\,(3+\sqrt{x+9})\right]$$

5. substitute x = 0

$$-2\,(3+3) = \boxed{-12}$$

Your turn · solve in your notebook · solutions at the back

1. **Ministry 2024/2025** Find the value of $\lim_{x\to 4} \dfrac{4-x}{5-\sqrt{x^2+9}}$.
2. **Book exercises 43–46** Given $\lim_{x\to a} f(x) = 2$, $\lim_{x\to a} g(x) = -3$, $\lim_{x\to a} h(x) = 0$, find (a) $\lim_{x\to a}[2f(x) - 3g(x)]$ (b) $\lim_{x\to a}[3f(x)g(x)]$ (c) $\lim_{x\to a} \dfrac{[f(x)]^2}{g(x)}$ (d) $\lim_{x\to a} \dfrac{2f(x)h(x)}{f(x)+h(x)}$.

6 Theorems: sin x / x

النهايات بالنظريات: النهاية المثلثية الخاصة

Ministry: 9 times in 8 years. You are given lim sin x / x = 1 — you do not prove it.

Make the inside of sin equal to the denominator, then use lim sin u / u = 1.

اجعل ما داخل sin هو المقام ثم استخدم sin u / u → 1.

Recipe الوصفة

1. **Make the inside of sin equal the denominator** — داخل sin = المقام
2. **Pull out the constants** — أخرج الثوابت
3. **Replace sin u / u by 1, then substitute** — استبدل ثم عوّض

Watch out

- You cannot cancel x from inside sin.
- Calculator in radians.

Tool. $\cos^2 x - 1 = -\sin^2 x$ · for $1 - \cos x$ multiply by $(1 + \cos x)$

Worked example Ministry 2022/2023

Given that $\lim_{x\to 0} \frac{\sin x}{x} = 1$, **evaluate** $\lim_{x\to 0} \frac{\cos^2 x - 1}{2x^2}$.

1. identity

$$\cos^2 x - 1 = -\sin^2 x$$

2. pull out the constant, show sin x / x

$$\lim_{x\to 0} \frac{-\sin^2 x}{2x^2} = \lim_{x\to 0} \left(-\frac{1}{2}\right)\left(\frac{\sin x}{x}\right)^2$$

3. use the given limit

$$-\frac{1}{2}(1)^2 = \boxed{-\frac{1}{2}}$$

Your turn · solve in your notebook · solutions at the back

1. Ministry 2016/2017 Find $\lim_{x\to 0} \frac{\sin 4x}{2x \cos x}$.
2. Ministry 2020/2021 Evaluate $\lim_{x\to 0} \frac{1 - \cos x}{x \sin x}$, if it exists.

7 The Squeeze Theorem

نظرية الشطيرة (الحصر)

Ministry: 3 different years, once as a written proof where each step earns a mark.

If f ≤ g ≤ h near a and lim f = lim h = L, then lim g = L.

إذا حُصرت الدالة بين دالتين لهما النهاية L فنهايتها L.

Recipe الوصفة

1. **Bound the wild part: −1 ≤ cos(1/x) ≤ 1** — قيّد الجزء المتذبذب
2. **Multiply through (x² ≥ 0 keeps the direction)** — اضرب
3. **Find both outer limits** — نهايتا الطرفين
4. **Write: by the Squeeze Theorem, limit = L** — اكتب النتيجة

Watch out

- Write both outer limits, then the theorem's name.
- Multiplying by a negative reverses the inequality.

Tool. |u| ≤ c means −c ≤ u ≤ c

Worked example Book example 3.8

Find $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$.

1. bound cosine

$$-1 \le \cos\left(\frac{1}{x}\right) \le 1 \quad (x \neq 0)$$

2. multiply by $x^2 \ge 0$ (direction stays)

$$-x^2 \le x^2 \cos\left(\frac{1}{x}\right) \le x^2$$

3. limits of both outer functions

$$\lim_{x\to 0}(-x^2) = 0, \qquad \lim_{x\to 0} x^2 = 0$$

4. Squeeze Theorem

$$\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right) = \boxed{0}$$

Your turn · solve in your notebook · solutions at the back

1. **Ministry 2023/2024 (paper part)** Suppose $|f(x)| \le M$ for all x. Use the Squeeze Theorem to prove that $\lim_{x\to 0} x^2 f(x) = 0$.
2. **Ministry style (2020/2021)** If $|g(x) - 3| \le 5\,|x - 1|$ for all x, find $\lim_{x\to 1} g(x)$.
3. **Book exercise 29** Find $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$.
4. **Ministry style** If $4 - x^2 \le f(x) \le 4 + x^2$ for all x, find $\lim_{x\to 0} f(x)$.

Objective 1 · The slope of a curve at a point ميل المنحنى عند نقطة

1.1 Method of example 1.1

Estimate the slope of $y = x^2 - 3x$ at $x = 1$, using the second points $x = 1.1,\ 1.01,\ 0.9,\ 0.99$.

1. the point

$f(1) = 1 - 3 = -2 \Rightarrow (1, -2)$

2. second points

$(1.1, -2.09),\ (1.01, -2.0099)$

$(0.9, -1.89),\ (0.99, -1.9899)$

3. m_sec on each side

$m_{sec} = -0.9,\ -0.99$ (right), $-1.1,\ -1.01$ (left)

4. estimate

slope $\approx \boxed{-1}$

1.2 Book exercise 2(b)

Estimate the slope of $y = x^3 + 2$ at $x = 2$, using the second points $x = 2.1,\ 2.01,\ 1.9,\ 1.99$.

1. the point

$f(2) = 10 \Rightarrow (2, 10)$

2. m_sec on each side

$$m_{sec} = \frac{y_2 - 10}{x_2 - 2} = 12.61,\ 12.0601 \text{ (right)}$$

$$= 11.41,\ 11.9401 \text{ (left)}$$

3. estimate

slope $\approx \boxed{12}$

Objective 2 · The length of a curve طول المنحنى

2.1 Ministry 2023/2024

Estimate the arc length of the curve $y = (x-1)^2$ for $0 \le x \le 3$ using $n = 2$ line segments.

1. x-values and points

$x = 0,\ 1.5,\ 3 \Rightarrow (0, 1),\ (1.5, 0.25),\ (3, 4)$

2. the two segments

$d_1 = \sqrt{1.5^2 + 0.75^2} \approx 1.6771, \quad d_2 = \sqrt{1.5^2 + 3.75^2} \approx 4.0389$

3. add

$L \approx \boxed{5.7159}$

2.2 Ministry 2025/2026

Estimate the arc length of the curve $y = \sin x$ on the interval $0 \le x \le \pi$ using $n = 2$ line segments.

1. three points

$(0, 0),\ \left(\frac{\pi}{2}, 1\right),\ (\pi, 0)$

2. two equal segments

$$L = 2\sqrt{\left(\frac{\pi}{2}\right)^2 + 1} \approx \boxed{3.7242}$$

2.3 Formative Assessment 1

Estimate the length of the curve $y = x^2 + 1$ for $-2 \le x \le 2$ using $n = 2$ line segments.

1. three points

$(-2, 5),\ (0, 1),\ (2, 5)$

2. two equal segments

$$L = 2\sqrt{2^2 + 4^2} = 2\sqrt{20} \approx \boxed{8.9443}$$

Objective 3 · The concept of a limit

مفهوم النهاية

3.1 Book exercise 3

Estimate $\lim_{x \to 2} \frac{x - 2}{x^2 - 4}$ with a table: $x = 1.9,\ 1.99,\ 2.01,\ 2.1$.

1. from the left

$f(1.9) \approx 0.2564, \quad f(1.99) \approx 0.2506$

2. from the right

$f(2.01) \approx 0.2494, \quad f(2.1) \approx 0.2439$

3. estimate

$$\approx \boxed{0.25 = \frac{1}{4}}$$

3.2 Method of example 2.1

Estimate $\lim_{x \to 1} \frac{x^3 - 1}{x - 1}$ with a table: $x = 0.9,\ 0.99,\ 1.01,\ 1.1$.

1. from the left

$f(0.9) = 2.71, \quad f(0.99) = 2.9701$

2. from the right

$f(1.01) = 3.0301, \quad f(1.1) = 3.31$

3. estimate

$$\approx \boxed{3}$$

Objective 4 · One-sided limits and existence

النهايات من جهة واحدة ووجود النهاية

4.1 Ministry 2021/2022

Evaluate $\lim_{x\to -1^+} \dfrac{|x+1|}{x^2-1}$.

1. for $x > -1$

$$|x+1| = x+1$$

2. factor and cancel

$$\lim_{x\to -1^+} \frac{x+1}{(x-1)(x+1)} = \lim_{x\to -1^+} \frac{1}{x-1}$$

3. substitute

$$\boxed{-\frac{1}{2}}$$

4.2 Ministry graph item

Use the graph to find: (a) $\lim_{x\to -2^-} f(x)$, **(b)** $\lim_{x\to -2^+} f(x)$, **(c)** $\lim_{x\to -2} f(x)$, **(d)** $\lim_{x\to 1} f(x)$, **(e)** $f(1)$.

1. at $x = -2$

(a) 2 (b) 0 (c) does not exist ($2 \neq 0$)

2. at $x = 1$

(d) 3 (e) $f(1) = 0$ (closed circle)

3. limit ≠ function value

$$\lim_{x\to 1} f(x) = 3 \neq f(1) = 0$$

4.3 Ministry 2019/2020 (paper part)

Find $\lim_{x\to 0} \dfrac{2x-|x|}{|3x|-2x}$**, if it exists.**

1. right side

$$\frac{2x-x}{3x-2x} = \frac{x}{x} = 1$$

2. left side

$$\frac{2x+x}{-3x-2x} = \frac{3x}{-5x} = -\frac{3}{5}$$

3. different → does not exist

$$1 \neq -\frac{3}{5} \Rightarrow \text{does not exist}$$

Objective 5 · Find a limit algebraically — إيجاد النهاية جبريًا

5.1 Ministry 2018/2019 (paper part)

Find $\lim_{x\to 2}\dfrac{x^3-8}{2x^2-3x-2}$**, if it exists.**

1. factor

$$\lim_{x\to 2}\frac{(x-2)(x^2+2x+4)}{(x-2)(2x+1)}$$

2. cancel, substitute

$$\lim_{x\to 2}\frac{x^2+2x+4}{2x+1}=\boxed{\frac{12}{5}}$$

5.2 Ministry 2020/2021

Evaluate $\lim_{x\to 0}\dfrac{\frac{1}{x+2}-\frac{1}{2}}{x}$**, if it exists.**

1. common denominator

$$\frac{1}{x+2}-\frac{1}{2}=\frac{2-(x+2)}{2(x+2)}=\frac{-x}{2(x+2)}$$

2. divide by x, cancel

$$\lim_{x\to 0}\frac{-x}{2(x+2)\,x}=\lim_{x\to 0}\frac{-1}{2(x+2)}$$

3. substitute

$$\boxed{-\frac{1}{4}}$$

5.3 Ministry 2019/2020

Find $\lim_{x\to 2}\left(\dfrac{1}{x-2}-\dfrac{4}{x^2-4}\right)$**, if it exists.**

1. common denominator, cancel

$$\frac{1}{x-2}-\frac{4}{x^2-4}=\frac{(x+2)-4}{(x-2)(x+2)}=\frac{x-2}{(x-2)(x+2)}=\frac{1}{x+2}$$

2. substitute

$$\lim_{x\to 2}\frac{1}{x+2}=\boxed{\frac{1}{4}}$$

Objective 6 · Theorems: rationalizing

النهايات بالنظريات: الضرب في المرافق

6a.1 Ministry 2024/2025

Find the value of $\lim_{x\to 4}\dfrac{4-x}{5-\sqrt{x^2+9}}$.

1. rationalize

$$\frac{(4-x)\,(5+\sqrt{x^2+9})}{25-(x^2+9)}=\frac{(4-x)\,(5+\sqrt{x^2+9})}{(4-x)(4+x)}$$

2. cancel, substitute

$$\lim_{x\to 4}\frac{5+\sqrt{x^2+9}}{4+x}=\frac{5+5}{8}=\boxed{\frac{5}{4}}$$

6a.2 Book exercises 43–46

Given $\lim_{x\to a} f(x)=2$, $\lim_{x\to a} g(x)=-3$, $\lim_{x\to a} h(x)=0$, **find (a)** $\lim_{x\to a}[2f(x)-3g(x)]$ **(b)** $\lim_{x\to a}[3f(x)g(x)]$ **(c)** $\lim_{x\to a}\dfrac{[f(x)]^2}{g(x)}$ **(d)** $\lim_{x\to a}\dfrac{2f(x)h(x)}{f(x)+h(x)}$.

1. (a) and (b)

$2(2)-3(-3)=\boxed{13}$, $\quad 3(2)(-3)=\boxed{-18}$

2. (c) and (d): denominators $\neq 0$

$$\frac{2^2}{-3}=\boxed{-\frac{4}{3}},\qquad \frac{2(2)(0)}{2+0}=\boxed{0}$$

Objective 6 · Theorems: sin x / x — النهايات بالنظريات: النهاية المثلثية الخاصة

6b.1 Ministry 2016/2017

Find $\lim_{x\to 0}\dfrac{\sin 4x}{2x\cos x}$.

1. inside of sin = denominator

$$\frac{\sin 4x}{2x\cos x}=\frac{4}{2}\cdot\frac{\sin 4x}{4x}\cdot\frac{1}{\cos x}$$

2. special limit, then x = 0

$$\to 2\cdot 1\cdot\frac{1}{\cos 0}=\boxed{2}$$

6b.2 Ministry 2020/2021

Evaluate $\lim_{x\to 0}\dfrac{1-\cos x}{x\sin x}$**, if it exists.**

1. × (1 + cos x)

$$\frac{(1-\cos x)(1+\cos x)}{x\sin x\,(1+\cos x)}=\frac{\sin^2 x}{x\sin x\,(1+\cos x)}=\frac{\sin x}{x\,(1+\cos x)}$$

2. special limit, substitute

$$\to 1\cdot\frac{1}{1+1}=\boxed{\frac{1}{2}}$$

7.1 Ministry 2023/2024 (paper part)

Suppose $|f(x)| \le M$ for all x. Use the Squeeze Theorem to prove that $\lim_{x\to 0} x^2 f(x) = 0$.

1. bound f

$$-M \le f(x) \le M$$

2. × $x^2 \ge 0$

$$-Mx^2 \le x^2 f(x) \le Mx^2$$

3. outer limits

$$\lim_{x\to 0}(-Mx^2) = 0 = \lim_{x\to 0} Mx^2$$

4. Squeeze Theorem

$$\lim_{x\to 0} x^2 f(x) = \boxed{0}$$

7.2 Ministry style (2020/2021)

If $|g(x) - 3| \le 5\,|x - 1|$ for all x, find $\lim_{x\to 1} g(x)$.

1. split $|\ldots|$

$$-5|x-1| \le g(x) - 3 \le 5|x-1|$$

2. add 3

$$3 - 5|x-1| \le g(x) \le 3 + 5|x-1|$$

3. outer limits → 3, Squeeze Theorem

$$\lim_{x\to 1} g(x) = \boxed{3}$$

7.3 Book exercise 29

Find $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$.

1. bound sine, × x^2

$$-x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le x^2$$

2. outer limits = 0, Squeeze Theorem

$$\boxed{0}$$

7.4 **Ministry style**

If $4 - x^2 \le f(x) \le 4 + x^2$ for all x, find $\lim_{x \to 0} f(x)$.

1. outer limits

$$\lim_{x \to 0}(4 - x^2) = 4 = \lim_{x \to 0}(4 + x^2)$$

2. Squeeze Theorem

$$\lim_{x \to 0} f(x) = \boxed{4}$$

Ready?

Tick each line · ضع علامة

- [ ] **I can estimate a slope from secant lines**
  أستطيع تقدير الميل بالقواطع
- [ ] **I can estimate an arc length with n = 2**
  أستطيع تقدير طول المنحنى
- [ ] **I can build a table and say what f(x) approaches**
  أستطيع بناء جدول القيم
- [ ] **I can read one-sided limits and say when a limit does not exist**
  أستطيع قراءة النهايات من جهة واحدة
- [ ] **I can factor, cancel and substitute**
  أستطيع التحليل والاختصار والتعويض
- [ ] **I can rationalize and use sin x / x**
  أستطيع الضرب في المرافق واستخدام sin x / x
- [ ] **I can use the Squeeze Theorem and write every step**
  أستطيع استخدام نظرية الشطيرة

On the day

Bring a scientific calculator. Radian mode for sin and cos. Write the first line at once: the first step already earns a mark.

Every limit

Substitute first. 0/0 → factor, rationalize, or use sin x / x. Keep the lim sign on every line until you substitute.

One mark for every written step

لكل خطوة تكتبها علامة